
Homework 6

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Note.

I frequently use the notation $f \Rightarrow g$ to denote a homotopy from f to g , because homotopy is close enough to a 2-morphism that I feel it's okay to use the notation. Also I think it can be viewed as a 2-morphism in an $(\infty, 1)$ -category, but that's beyond me at the moment. Also note that in a previous homework, I essentially proved the universal property of products in **Top**, and I have been quietly referencing it throughout the semester. I cannot be asked to continuously say "this was proven in a previous homework" every time, so I will be assuming it throughout. I also assume the universal property of quotients in **Top**, because it's easy to show.

Notation. If $f : A \rightarrow X$ and $g : B \rightarrow Y$ are morphisms, then the morphism $f \times g$ will denote the morphism induced by the universal property of products. If $f : A \rightarrow X$ and $g : B \rightarrow Y$ are morphisms now with *different* domains, then the morphism $f \times g$ is the morphism induced by the product bifunctor; equivalently, it is the induced morphism in the diagram

$$\begin{array}{ccccc} & & A \times B & & \\ & f \circ \pi_A \swarrow & \downarrow f \times g & \searrow g \circ \pi_B & \\ X & \xleftarrow{\pi_X} & X \times Y & \xrightarrow{\pi_Y} & Y \end{array}$$

Problem 1

Due to the notation used in the problem statement, I will assume X is path-connected since π_1 was stated without specifying a base point. This, together with the fact that D^2 is path-connected, makes almost all mention of base points immaterial in this problem. At some points when using constant maps it can be implicitly assumed what the base point is.

First, note that loops in X can be equivalently viewed as morphisms $S^1 \rightarrow X$. To see this explicitly, note that every loop $\varphi : [0, 1] \rightarrow X$ has $\varphi(0) = \varphi(1)$, and thus it factors uniquely through the quotient

$$\begin{array}{ccc} [0, 1] & \xrightarrow{\varphi} & X \\ \pi \downarrow & \nearrow \bar{\varphi} & \\ [0, 1] / \sim & & \end{array}$$

for π the canonical projection, where $\sim$ is the relation $0 \sim 1$. It's clear that $[0, 1] / \sim \cong S^1$, so every loop φ can be equivalently viewed as a continuous map $\bar{\varphi} : S^1 \rightarrow X$. To fully formalize this idea up as far as will be needed for this problem, consider the following lemma, which will implicitly be invoked throughout the problem.

Lemma 1. *With notation as above, φ is null-homotopic if and only if $\bar{\varphi}$ is null-homotopic.*

Proof. First, let φ be null-homotopic to the constant map x_0 . Since φ is a loop, $\varphi(0) = \varphi(1)$. Let $\Phi : \varphi \Rightarrow x_0$ be the homotopy in question, and define the map

$$\Psi := \Phi(x).$$

This is well defined; that is, for any $x \in [0, 1]$, we can take $\Phi(x, 0) = \Phi(x, 1)$ without loss of generality. This assumption follows from the fact that for any x , we can simply concatenate on the left and right by paths that flow from some equivalent start and end points to the start and end points of $\Phi(x, -)$. With this definition, it is exceedingly clear that Φ now defines a homotopy $\bar{\varphi} \Rightarrow x_0$.

For the converse, let $\bar{\varphi}$ be null-homotopic to the constant map x_0 . Since π is homotopic to itself by the identity homotopy 1_π , we have a diagram

$$\begin{array}{ccccc} [0, 1] & \xrightarrow{\pi} & & \xrightarrow{\bar{\varphi}} & X \\ & \Downarrow & [0, 1] / \sim & \Downarrow & \\ & \pi & & x_0 & \end{array}$$

By lemma 2 (yes, I presented the results out of order. Cope), there is an induced homotopy

$$\varphi = \bar{\varphi} \circ \pi \Rightarrow x_0 \circ \pi = x_0,$$

and thus φ is null-homotopic. ■

Suppose that every morphism $f : S^1 \rightarrow X$ extends to a morphism $\bar{f} : D^2 \rightarrow X$; in other words, the diagram

$$\begin{array}{ccc} D^2 & \xrightarrow{\bar{f}} & X \\ \uparrow \iota & \nearrow f & \\ S^1 & & \end{array}$$

commutes, with ι the inclusion $S^1 \hookrightarrow D^2$. This means that every loop f can be viewed as the composition $\bar{f} \circ \iota$. However, note that ι is a morphism $S^1 \rightarrow D^2$, and thus a loop in D^2 . Using the fact that $\pi_1(D^2) = 0$, we have that all loops in D^2 are null-homotopic, and thus there is a homotopy $\Phi : [0, 1] \times S^1 \rightarrow D^2$ with $\Phi(0, -) = \iota$ and $\Phi(1, -) = x_0$, for some fixed $x_0 \in D^2$. Additionally, note that $\bar{f}$ is homotopic to itself by the identity homotopy $1_{\bar{f}} : [0, 1] \times D^2 \rightarrow X$, defined as $1_{\bar{f}}(x, -) = \bar{f}$ for all $x \in D^2$. We thus have homotopies

$$\begin{array}{ccccc} S^1 & \xrightarrow{\iota} & D^2 & \xrightarrow{\bar{f}} & X \\ & \Downarrow & & \Downarrow & \\ & x_0 & & \bar{f} & \end{array}$$

Consider the following lemma.

Lemma 2. *Homotopies $\Phi_1 : f_1 \Rightarrow g_1$ and $\Phi_2 : f_2 \Rightarrow g_2$ can be composed horizontally; that is, if $g_1 \sim f_1$ and $g_2 \sim f_2$ for $\sim$ the homotopy relation, then $g_2 \circ g_1 \sim f_2 \circ f_1$.*

Proof. Let

$$\begin{array}{ccccc} X & \xrightarrow{f_1} & Y & \xrightarrow{g_1} & Z, \\ X & \xrightarrow{f_2} & Y & \xrightarrow{g_2} & Z \end{array}$$

be morphisms, and let $\Phi_1 : [0, 1] \times X \rightarrow Y$ and $\Phi_2 : [0, 1] \times Y \rightarrow Z$ be homotopies $f_1 \Rightarrow g_1$ and $f_2 \Rightarrow g_2$, respectively. Define the horizontal composite $\Phi_2 \cdot \Phi_1 : [0, 1] \times X \rightarrow Z$ as

$$(\Phi_2 \cdot \Phi_1)(x, -) = \Phi_2(x, \Phi_1(x, -)).$$

To show that this is a homotopy $f_2 \circ f_1 \Rightarrow g_2 \circ g_1$, first note that

$$\Phi_2(0, \Phi_1(0, -)) = \Phi_2(0, f_1(-)) = f_2(f_1(-)) = f_2 \circ f_1,$$

by definition of Φ_1 and Φ_2 . Similarly,

$$\Phi_2(1, \Phi_1(1, -)) = \Phi_2(1, g_1(-)) = g_2 \circ g_1.$$

Finally, to show $\Phi_2 \cdot \Phi_1$ is continuous, note that we can view $\Phi_2 \cdot \Phi_1$ as the composition $\Phi_2 \circ (\pi \times \Phi_1)$, where π is the natural projection $[0, 1] \times X \rightarrow [0, 1]$, and $\pi \times \Phi_1$ is the map induced by the universal property of products

$$\begin{array}{ccccc} & & [0, 1] \times X & & \\ & \swarrow \pi & \downarrow \pi \times \Phi_1 & \searrow \Phi_1 & \\ [0, 1] & \longleftarrow & [0, 1] \times Y & \longrightarrow & Y \end{array}$$

Proof of this is immediate:

$$(\Phi_2 \circ (\pi \times \Phi_1))(x, -) = \Phi_2(x, \Phi_1(x, -)) = (\Phi_2 \cdot \Phi_1)(x, -).$$

Since π and Φ_1 are continuous, so is $\pi \times \Phi_1$, and thus so is $\Phi_2 \circ (\pi \times \Phi_1) = \Phi_2 \cdot \Phi_1$. Therefore, horizontal composition of homotopies holds, and $f_2 \circ f_1 \sim g_2 \circ g_1$. $\blacksquare$

By lemma 2, there is a homotopy $\bar{f} \circ \iota \Rightarrow \bar{f} \circ x_0$. Since the composition $\bar{f} \circ x_0$ is clearly still a constant map $x \mapsto \bar{f}(x_0)$, and since $\bar{f} \circ \iota$ is homotopic to $\bar{f} \circ x_0$, we have that $\bar{f} \circ \iota$ is null-homotopic. Since any loop f is equal to such a composition $\bar{f} \circ \iota$ by assumption, we have that every loop f is null-homotopic, and thus $\pi_1(X) \cong 0$.

Now for the converse, let $\pi_1(X) \cong 0$, and let $f : S^1 \rightarrow X$ be continuous. Since f can be viewed as a loop, it is necessarily null-homotopic $\Phi : f \Rightarrow x_0$ for x_0 the constant map $x \mapsto x_0$ with x_0 fixed.

Let $(x_1, y_1) \sim (x_2, y_2)$ iff $x_1 = x_2 = 1$ be a relation on $[0, 1] \times S^1$, and note that since $\Phi(1, -) = x_0$, by the universal property of quotients it factors uniquely through the quotient:

$$\begin{array}{ccc} [0, 1] \times S^1 & \xrightarrow{\Phi} & X \\ \downarrow & \nearrow & \\ [0, 1] \times S^1 / \sim & & \end{array}$$

For simplicity, define $A := [0, 1] \times S^1 / \sim$. We claim that $A \cong D^2$. Note that every element of A is of the form $(x, \cos \theta, \sin \theta)$ for some $x \in [0, 1]$ and $\theta \in [0, 2\pi)$. Let $\psi : [0, 1] \times S^1 \rightarrow D^2$ be the map

$$(x, \cos \theta, \sin \theta) \mapsto (1 - x)(\cos \theta, \sin \theta).$$

Since $\psi(1, x, y) = 0$, this map identifies all points with $x = 1$, it also factors through A and induces a continuous map $\bar{\psi} : A \rightarrow D^2$ by mapping $[a] \mapsto \psi(a)$.

Now we show that ψ (not $\bar{\psi}$) is surjective. Given $r(\cos \theta, \sin \theta) \in D^2$, we have that since $r \in [0, 1]$, it can be viewed as $(1 - x)$ for some $x \in [0, 1]$. It is then manifestly surjective, as for every such x , there exists $(x, \cos \theta, \sin \theta) \in [0, 1] \times S^1$. Also note that the map is not injective. However, for any $(x, \cos \theta, \sin \theta) \neq (y, \cos \phi, \sin \phi)$, it can easily be shown that

$$(1 - x)(\cos \theta, \sin \theta) = (1 - y)(\cos \phi, \sin \phi) \implies x = y = 1$$

simply by our knowledge of S^1 and D^2 . Therefore, $\psi(x, \cos \theta, \sin \theta) = \psi(y, \cos \phi, \sin \phi)$ if and only if $x = y = 1$. The quotienting relation $\sim$ defining A is thus equivalent to the relation $x \sim y$ iff $\psi(x) = \psi(y)$. The canonical factorization of ψ is precisely its factorization through A :

$$\begin{array}{ccccc} & & \psi & & \\ & \curvearrowright & & \searrow & \\ [0, 1] \times S^1 & \xrightarrow{\pi} & A & \xrightarrow[\overline{\psi}]{\text{---}} & \text{im } \psi \xrightarrow{=} D^2, \end{array}$$

where the fact that $\text{im } \psi = D^2$ follows from the surjectivity of ψ , and thus $\overline{\psi}$ is a bijection $A \rightarrow D^2$. Consider the following lemma.

Lemma 3. *Let X be compact. Then the quotient $X/\sim$ is compact for any relation $\sim$.*

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of $X/\sim$. Since the quotient topology is defined such that the canonical projection $\pi : X \rightarrow X/\sim$ is continuous, we have that $\pi^{-1}(U_\alpha)$ is open in X for all α , and thus

$$\bigcup_{\alpha \in A} \pi^{-1}(U_\alpha) = X$$

is an open cover. Since X is compact, this admits a finite subcover

$$\bigcup_{i=1}^n \pi^{-1}(U_i) = X,$$

and since π is manifestly surjective, we have

$$X/\sim = \pi(X) = \pi\left(\bigcup_{i=1}^n \pi^{-1}(U_i)\right) = \bigcup_{i=1}^n \pi(\pi^{-1}(U_i)) = \bigcup_{i=1}^n U_i.$$

Therefore, $X/\sim$ is compact. ■

By lemma 3, A is compact. Since D^2 is Hausdorff, we have that $\overline{\psi}$ is a continuous bijection from a compact space to a Hausdorff space. By a result I proved in the last homework, $\overline{\psi}$ is thus a homeomorphism, and $A \cong D^2$.

Returning to our discussion of the homotopy Φ , we thus have that the diagram

$$\begin{array}{ccc} [0, 1] \times S^1 & \xrightarrow{\Phi} & X \\ \downarrow & \nearrow \varphi & \\ D^2 & & \end{array}$$

commutes. We claim that φ is an extension of f to D^2 . For this to be true, we would need the diagram

$$\begin{array}{ccc} D^2 & & \\ \uparrow \iota & \searrow \varphi & \\ S^1 & \xrightarrow{f} & X \end{array}$$

to commute. Note that $\Phi(0, -) = f$ since Φ is a homotopy $f \Rightarrow x_0$. Also note that due to the definition of the homeomorphism $\overline{\psi}$, points $(0, x) \in [0, 1] \times S^1$ are identified with the boundary of

D^2 . In other words, the equivalence class of $(0, \cos \theta, \sin \theta)$ corresponds to $(\cos \theta, \sin \theta)$. We thus have that any point $x \in D^2$ with radius 1 must have

$$\varphi(x) = \Phi(0, [x]) = \Phi(0, x) = f(x).$$

Since $\iota(x)$ has radius 1 for any $x \in S^1$, we have

$$(\varphi \circ \iota)(x) = f(x),$$

and thus $\varphi \circ \iota = f$. Therefore, the diagram commutes, and the statement is proven. ■

Problem 2

Notation. In this problem, $\bullet$ will denote loop concatenation, $\circ$ will denote function composition, and juxtaposition of functions will denote pointwise group multiplication. In general, 1_X denotes an identity morphism on X , and e_G denotes the identity of a group G .

Let G be a path connected topological group, and since π_1 is invariant under choice of base point for path connected spaces, we can choose $\pi_1(G)$ to denote $\pi_1(G, e_G)$ without loss of generality.

Let α, β be loops based at e_G . We will first show $\alpha\beta$ is a loop. Let $\Delta : [0, 1] \rightarrow [0, 1] \times [0, 1]$ be the map $x \mapsto (x, x)$, and let $\mu : G \times G \rightarrow G$ be group multiplication. We are given that μ is continuous, and obviously $\Delta = 1_{[0,1]} \times 1_{[0,1]}$, so Δ is also continuous. The identity function 1_G is clearly continuous, as it is the identity morphism in both **Top** and **Grp**, and thus we have that $\alpha \times 1_{[0,1]}$ and $1_G \times \beta$ are continuous. Therefore, we can represent the function $\alpha\beta$ as the composition of continuous functions (where 1 is a shorthand for the relevant identity)

$$\begin{array}{c} \alpha\beta \\ \curvearrowright \\ [0, 1] \xrightarrow{\Delta} [0, 1] \times [0, 1] \xrightarrow{\alpha \times 1} G \times [0, 1] \xrightarrow{1 \times \beta} G \times G \xrightarrow{\mu} G \end{array}$$

and thus $\alpha\beta : [0, 1] \rightarrow G$ is continuous.

Next, note that since α, β are both loops based at e_G , we have that

$$\alpha(0)\beta(0) = e_G^2 = e_G, \quad \alpha(1)\beta(1) = e_G^2 = e_G$$

Therefore, $\alpha\beta$ is also a loop based at e_G .

Let $\alpha_1, \alpha_2, \beta_1, \beta_2$ be loops based at e_G . Note that for all $0 \leq x \leq 1/2$, we have

$$(\alpha_1 \bullet \beta_1)(x) \cdot (\alpha_2 \bullet \beta_2)(x) = \alpha_1(2x) \cdot \beta_1(2x).$$

Similarly, for $1/2 \leq x \leq 1$, we have

$$(\alpha_1 \bullet \beta_1)(x) \cdot (\alpha_2 \bullet \beta_2)(x) = \alpha_2(2x - 1) \cdot \beta_2(2x - 1).$$

Note also that for $0 \leq x \leq 1/2$, we have

$$(\alpha_1\beta_1) \bullet (\alpha_2\beta_2)(2x) = (\alpha_1\beta_1)(x) = \alpha_1(2x)\beta_1(2x),$$

and for $1/2 \leq x \leq 1$, we have

$$(\alpha_1\beta_1) \bullet (\alpha_2\beta_2)(2x) = (\alpha_2\beta_2)(x) = \alpha_2(2x - 1)\beta_2(2x - 1),$$

Since $\alpha_i \beta_i$ is a loop for all i , we thus have

$$(\alpha_1 \bullet \beta_1)(\alpha_2 \bullet \beta_2) = (\alpha_1 \beta_1) \bullet (\alpha_2 \beta_2).$$

In particular,

$$(\alpha \bullet e)(e \bullet \beta) = (\alpha e) \bullet (e \beta),$$

where e in this case denotes the constant loop at e_G . We trivially have

$$(\alpha e) \bullet (e \beta) = \alpha \bullet \beta,$$

and we also trivially have that

$$\alpha \bullet e \sim \alpha, \quad e \bullet \beta \sim \beta$$

for $\sim$ homotopy, since we already have e is an identity up to homotopy with respect to loop concatenation, simply by our investigation of π_1 .

Next, note that since we have homotopies

$$\alpha \bullet e \Rightarrow \alpha, \quad e \bullet \beta \Rightarrow \beta,$$

and we have a similar decomposition (where once again 1 is a shorthand for the relevant identity)

$$\begin{array}{c}
 \alpha \bullet \beta \\
 \curvearrowright \\
 [0, 1] \xrightarrow{\Delta} [0, 1] \times [0, 1] \xrightarrow{(\alpha \bullet e) \times 1} G \times [0, 1] \xrightarrow{1 \times (\beta \bullet e)} G \times G \xrightarrow{\mu} G. \\
 \curvearrowleft \\
 (\alpha \bullet e)(e \bullet \beta)
 \end{array}$$

Lemma 4. *Let*

$$\begin{array}{ccc}
 & X & \\
 f_1 \swarrow & & \searrow f_2 \\
 Y & & Z \\
 g_1 \nwarrow & & \nearrow g_2
 \end{array}$$

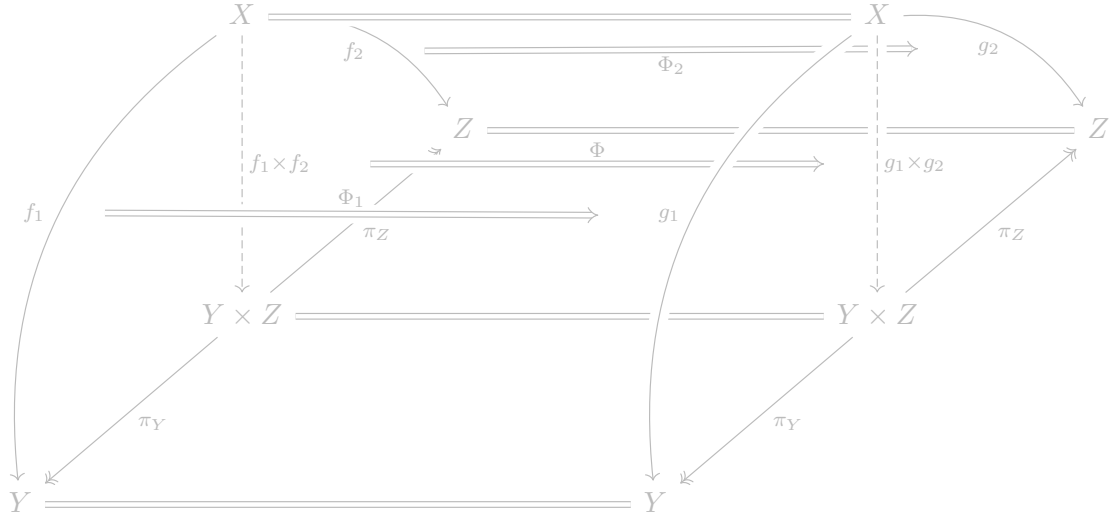
be continuous maps, and let

$$\Phi_1 : f_1 \Rightarrow g_1, \quad \Phi_2 : f_2 \Rightarrow g_2$$

be homotopies. Then there is an induced homotopy

$$\Phi : f_1 \times f_2 \Rightarrow g_1 \times g_2.$$

Proof. This is equivalent to showing there exists a homotopy Φ making the diagram



commute up to the given homotopies. This diagram does not help the proof at all I just thought it looked cool. Since the homotopies are morphisms

$$\begin{array}{ccc} & [0, 1] \times X & \\ \Phi_1 \swarrow & & \searrow \Phi_2 \\ Y & & Z \end{array},$$

we can consider the product $\Phi_1 \times \Phi_2 : [0, 1] \times X \times [0, 1] \times X \rightarrow Y \times Z$, defined as

$$(\Phi_1 \times \Phi_2)(t_1, x, t_2, y) = (\Phi_1(t_1, x), \Phi_2(t_2, y)).$$

Define $\Phi : [0, 1] \times X \times X \rightarrow Y \times Z$ as the composition

$$\begin{array}{c} \Phi \\ \curvearrowright \\ [0, 1] \times X \times X \xrightarrow{\Delta \times 1_X \times 1_X} [0, 1] \times [0, 1] \times X \times X \xrightarrow{\gamma} [0, 1] \times X \times [0, 1] \times X \xrightarrow{\Phi_1 \times \Phi_2} Y \times Z, \end{array}$$

where $\Delta : [0, 1] \rightarrow [0, 1] \times [0, 1]$ is still the map $t \mapsto (t, t)$, and γ is the commutator

$$\gamma(t_1, t_2, x, y) = (t_1, x, t_2, y).$$

More precisely, γ is the relevant component of the natural isomorphism $_ \times _ \Rightarrow _ \times _$ which gives commutivity of products.

Since $\Phi_1 \times \Phi_2$ is the product of continuous functions, it is continuous. Similarly, $\Delta \times 1_X \times 1_X$ is continuous. Finally, γ is trivially shown to be continuous, so we have that Φ is a composition of continuous functions, and thus continuous. It suffices to show Φ determines a homotopy $f_1 \times f_2 \Rightarrow$

$g_1 \times g_2$. Note that we can explicitly describe $\Phi(t, x, y)$ as

$$\begin{aligned} ((\Phi_1 \times \Phi_2) \circ \gamma \circ (\Delta \times 1_X \times 1_X))(t, x, y) &= ((\Phi_1 \times \Phi_2) \circ \gamma)(t, t, x, y) \\ &= (\Phi_1 \times \Phi_2)(t, x, t, y) \\ &= (\Phi_1(t, x), \Phi_2(t, y)). \end{aligned}$$

We thus have

$$\Phi(0, x, y) = (\Phi_1(0, x), \Phi_2(0, y)) = (f_1(x), f_2(y)) = (f_1 \times f_2)(x, y).$$

Similarly,

$$\Phi(1, x, y) = (\Phi_1(1, x), \Phi_2(1, y)) = (g_1(x), g_2(y)) = (g_1 \times g_2)(x, y).$$

Therefore, there exists a homotopy $\Phi : f_1 \times f_2 \Rightarrow g_1 \times g_2$. ■

Recall that there exist homotopies

$$\alpha \bullet e \Rightarrow \alpha, \quad e \bullet \beta \Rightarrow \beta.$$

Since there exists an identity homotopy $1_{[0,1]} : 1_{[0,1]} \Rightarrow 1_{[0,1]}$, by lemma 4, there exist homotopies

$$\Phi_\alpha : (\alpha \bullet e) \times 1_{[0,1]} \Rightarrow \alpha \times 1_{[0,1]}, \quad \Phi_\beta : 1_{[0,1]} \times (e \bullet \beta) \Rightarrow 1_{[0,1]} \times \beta.$$

Since there also exist identity homotopies $1_\Delta : \Delta \Rightarrow \Delta$ and $1_\mu : \mu \Rightarrow \mu$, it follows that the diagram

$$\begin{array}{ccccccc} & & & \alpha \bullet \beta & & & \\ & \swarrow & & \searrow & & \swarrow & \\ [0, 1] & \xrightarrow{\Delta} & [0, 1] \times [0, 1] & \xrightarrow{(\alpha \bullet e) \times 1} & G \times [0, 1] & \xrightarrow{1 \times (\beta \bullet e)} & G \times G \xrightarrow{\mu} G \\ \parallel & \Downarrow 1_\Delta & \parallel & \Downarrow \Phi_\alpha & \parallel & \Downarrow \Phi_\beta & \parallel & \Downarrow 1_\mu & \parallel \\ [0, 1] & \xrightarrow{\Delta} & [0, 1] \times [0, 1] & \xrightarrow{\alpha \times 1} & G \times [0, 1] & \xrightarrow{1 \times \beta} & G \times G \xrightarrow{\mu} G \\ & \searrow & & \swarrow & & \searrow & \\ & & & \alpha \beta & & & \end{array}$$

commutes up to the given homotopies, where once again 1 is a shorthand for $1_{[0,1]}$. By repeated applications of lemma 2, there exists a homotopy

$$\alpha \bullet \beta \Rightarrow \alpha \beta.$$

Since homotopies are invertible, loop concatenation in G is precisely group multiplication of loops in G , up to homotopy, and thus

$$[\alpha] \cdot [\beta] = [\alpha \bullet \beta] = [\alpha \beta]$$

in $\pi_1(G)$.

This whole mess will be useful soon. For now, consider the map

$$H : [0, 1] \times [0, 1] \rightarrow G, \quad (x, y) \mapsto \alpha(x)\beta(y).$$

Note that $H(0,0) = \alpha(0) = e_G$, and thus $H \circ 0 = e$. Since H is homotopic to itself by the identity homotopy, we once again have a diagram

$$\begin{array}{ccccc}
 & & F & & \\
 & \nearrow & & \searrow & \\
 [0,1] & \xrightarrow{d} & [0,1] \times [0,1] & \xrightarrow{H} & G \\
 \parallel & \Downarrow \Phi & \parallel & \Downarrow 1_H & \parallel \\
 [0,1] & \xrightarrow{0} & [0,1] \times [0,1] & \xrightarrow{H} & G \\
 & \searrow & & \nearrow & \\
 & & e & &
 \end{array}$$

which commutes up to the given homotopies. By lemma 2, there is an induced homotopy

$$\Psi : F \Rightarrow e.$$

We thus have that φ is homotopic to F , which is null-homotopic. Since homotopy is an equivalence relation, we have that φ is null-homotopic, and thus

$$\alpha\beta\alpha^{-1}\beta^{-1} \sim e.$$

Since $\alpha\beta\alpha^{-1}\beta^{-1}$ is homotopic to $\alpha \bullet \beta \bullet \alpha^{-1} \bullet \beta^{-1}$, we can pull it into $\pi_1(G)$ to observe that

$$[\alpha][\beta][\alpha]^{-1}[\beta]^{-1} = [e] = e_{\pi_1(G)} \implies [\alpha][\beta] = [\beta][\alpha]$$

by our knowledge of group theory. Therefore, $\pi_1(G)$ is abelian. ■

Problem 3

Given a morphism $f : (X, x_0) \rightarrow (Y, y_0)$, define the induced morphism $f_* := \pi_1(f)$. let $\text{id}_X : (X, x_0) \rightarrow (X, x_0)$ be the identity morphism, and note that due to our prior knowledge of the category **Grp**, the identity morphism id_G on any group G is the map $g \mapsto g$. It thus suffices to show id_{X*} is the map $[\varphi] \mapsto [\varphi]$. By definition,

$$\text{id}_{X*}[\varphi] = [\text{id}_X \circ \varphi] = [\varphi],$$

since id_X is an identity morphism in **Top***, and the homotopy class of φ consists entirely of morphisms in **Top***. Therefore, id_{X*} is the identity morphism $\text{id}_{\pi_1(X, x_0)}$.

Let

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

be morphisms in **Top**, for X, Y, Z path-connected spaces (such that π_1 is well-defined up to isomorphism for any choice of base point). We have

$$(g_* \circ f_*)[\varphi] = g_*[f \circ \varphi] = [g \circ (f \circ \varphi)] = [(g \circ f) \circ \varphi] = (g \circ f)_*[\varphi].$$

Therefore, π_1 preserves morphism composition, and thus π_1 is a functor (if we restrict **Top** to path connected spaces; I'm pretty sure the proof for functorality **Top*** $\rightarrow$ **Grp** is the same but that wasn't the problem soooooo).

Since π_1 is a functor, if $f : X \rightarrow Y$ is a homeomorphism, then it has a two-sided continuous inverse $f^{-1} : Y \rightarrow X$. We thus have

$$\pi_1(f) \circ \pi_1(f^{-1}) = \pi_1(f \circ f^{-1}) = \pi_1(\text{id}_Y) = \text{id}_{\pi_1(Y)},$$

and similarly $\pi_1(f^{-1}) \circ \pi_1(f) = \text{id}_{\pi_1(X)}$. Therefore, $\pi_1(X) \cong \pi_1(Y)$. ■

Problem 4

We showed in class that $S^n \setminus \{*\} \cong \mathbb{R}^n$ by stereographic projection. Let $f : X \rightarrow S^n$ be a non-surjective continuous map. It follows that there exists $*$ $\in S^n \setminus \text{im } f$. Composing with the stereographic projection $s : S^n \setminus \{*\} \rightarrow \mathbb{R}^n$, we have a continuous map

$$F : X \xrightarrow{f|_{\text{im } f}} \text{im } f \xrightarrow{\iota} S^n \setminus \{*\} \xrightarrow{s} \mathbb{R}^n,$$

where ι is the inclusion map. Note that we can define a homotopy

$$\Phi : [0, 1] \times X \rightarrow \mathbb{R}^n, \quad (t, x) \mapsto tF(x).$$

We clearly have $\Phi(0, -) = 0$ and $\Phi(1, -) = F$. Moreover, since this is a product of continuous functions, it is indeed continuous by an immediate generalization of a proof given in a previous homework, and thus Φ is indeed a homotopy $0 \Rightarrow F$. Thus, F is null-homotopic. Note that since s is a homeomorphism, it has an inverse s^{-1} , and by definition of F , we have

$$s^{-1} \circ F = \iota \circ f|_{\text{im } f}.$$

Composing with the inclusion $\iota_S : S^n \setminus \{*\} \hookrightarrow S^n$ gives us

$$\iota_S \circ \iota \circ f|_{\text{im } f} = f.$$

Also that

$$(\iota_S \circ s^{-1} \circ 0)(x) = (\iota_S \circ s^{-1})(0)$$

is clearly still a constant map. We will abuse notation slightly and call the composition map 0 as well. Since ι_S and s^{-1} are homotopic to themselves by the identity homotopies, we have the diagram

$$\begin{array}{ccccccc} & & & f & & & \\ & \nearrow & & & \searrow & & \\ X & \xrightarrow{F} & \mathbb{R}^n & \xrightarrow{s^{-1}} & S^n \setminus \{*\} & \xrightarrow{\iota_S} & S^n \\ \parallel & \downarrow \Phi & \parallel & \downarrow 1_{s^{-1}} & \parallel & \downarrow 1_{\iota_S} & \parallel \\ X & \xrightarrow{0} & \mathbb{R}^n & \xrightarrow{s^{-1}} & S^n \setminus \{*\} & \xrightarrow{\iota_S} & S^n \\ & \searrow & & & \nearrow & & \\ & & & 0 & & & \end{array}$$

commutes up to the given homotopies. By repeated applications of lemma 2, there is an induced homotopy

$$f \Rightarrow 0.$$

Therefore, f is null-homotopic. ■